



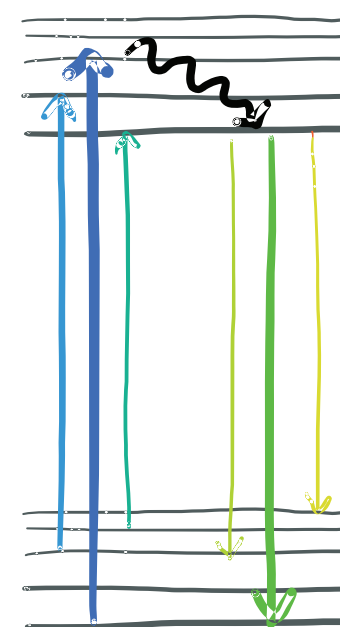
Fitting Convolved Data by Markov Chain Monte Carlo Likelihood Maximization

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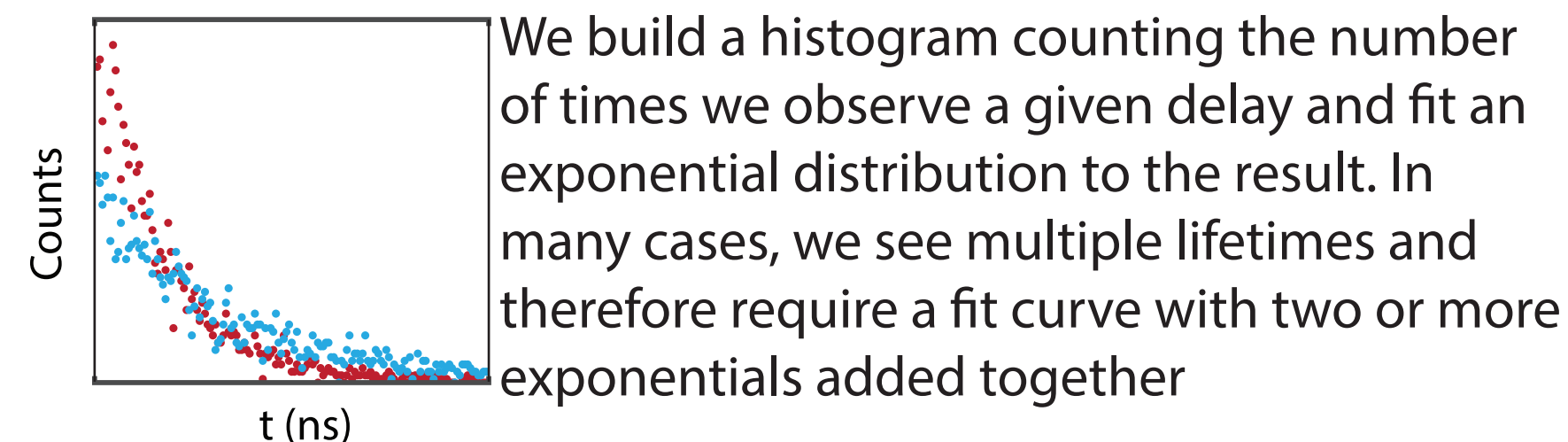


Background

When a molecule is excited with a photon of the appropriate wavelength, an electron is excited into a higher energy level. After a few nanoseconds, the molecule reconfigures itself and the electron relaxes, emitting a lower-energy photon

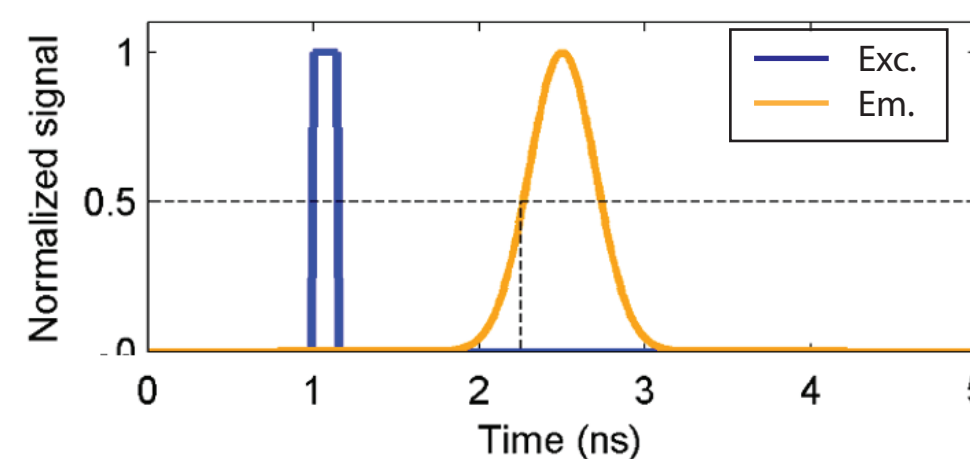


By measuring the nanoseconds-long delay between excitation and emission, we can identify unique molecules or make measurements of the environment around the fluorescent molecule.



We build a histogram counting the number of times we observe a given delay and fit an exponential distribution to the result. In many cases, we see multiple lifetimes and therefore require a fit curve with two or more exponentials added together

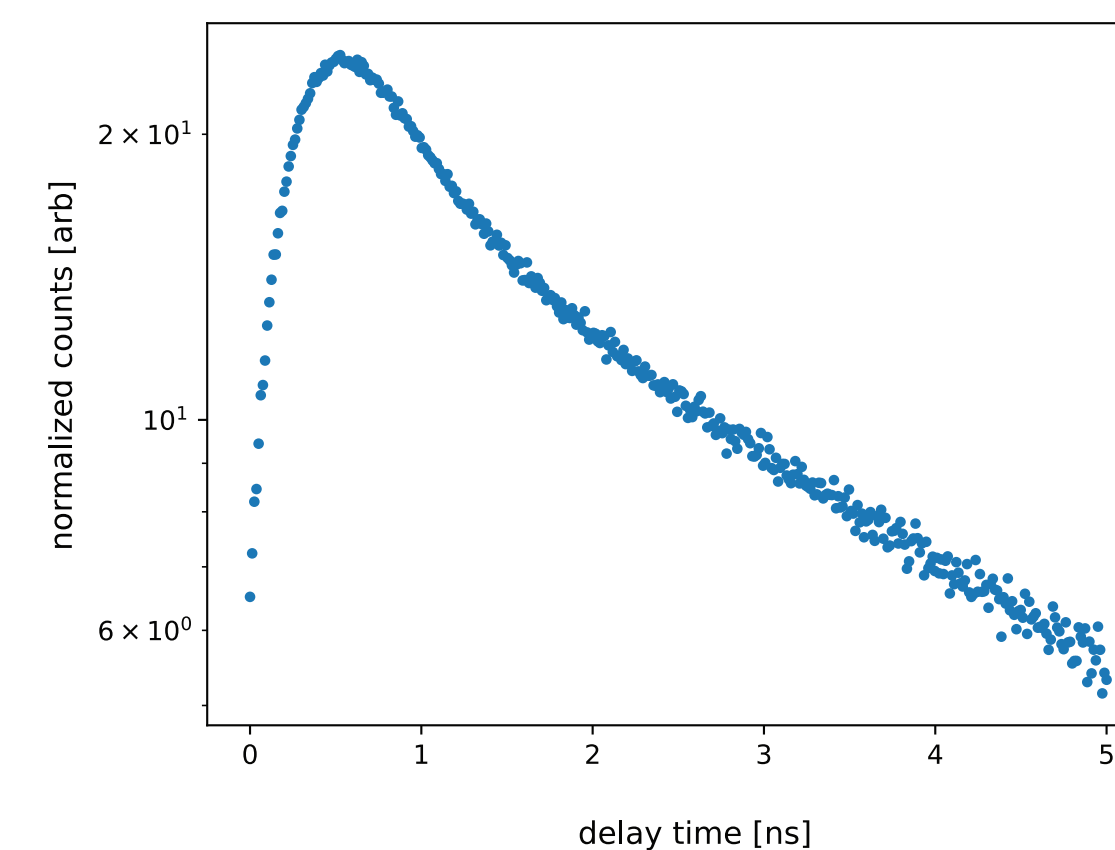
However, the detector we use to measure incoming emission photons isn't perfect- each incoming photon registers not as a pulse but as a broad peak called the Instrument Response Function (IRF). The overall histogram we measure is the true histogram of delay counts, convolved with the IRF. We therefore cannot simply fit a (double-) exponential to our observed histogram.



GUIDING QUESTION

How can we fit a curve to data that has been convolved with a known convolution kernel?

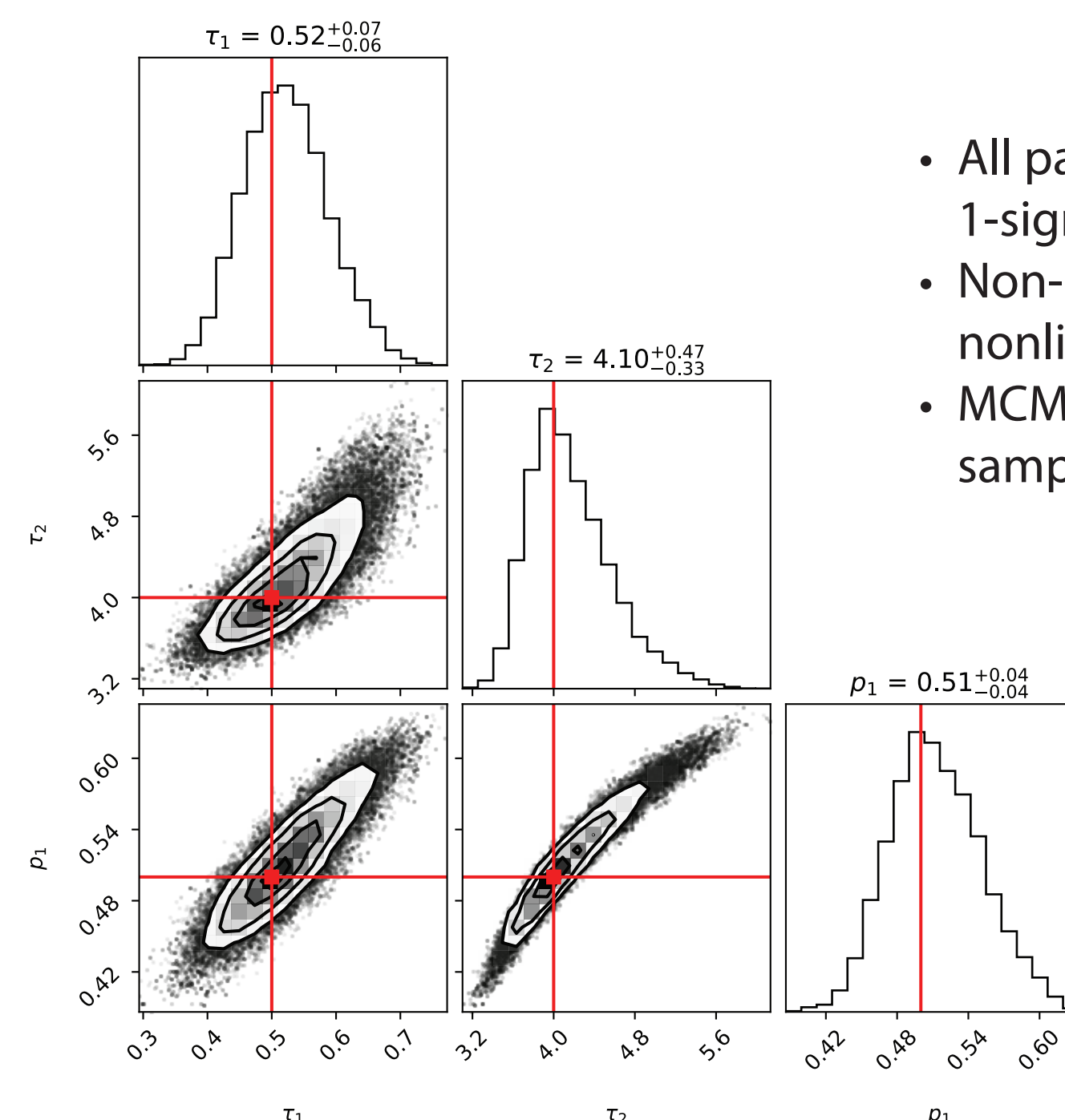
Results



- Simulated double exponential decay with lifetimes 0.5, 4ns added in equal measure
$$y = p_1 e^{-t/\tau_1} + (1 - p_1) e^{-t/\tau_2}$$
$$\tau_1 = 0.5\text{ns}, \tau_2 = 4\text{ns}, p = 0.5$$
- Convolved with gaussian with width 0.5ns
- Added noise with standard deviation 0.2

MCMC Parameters:

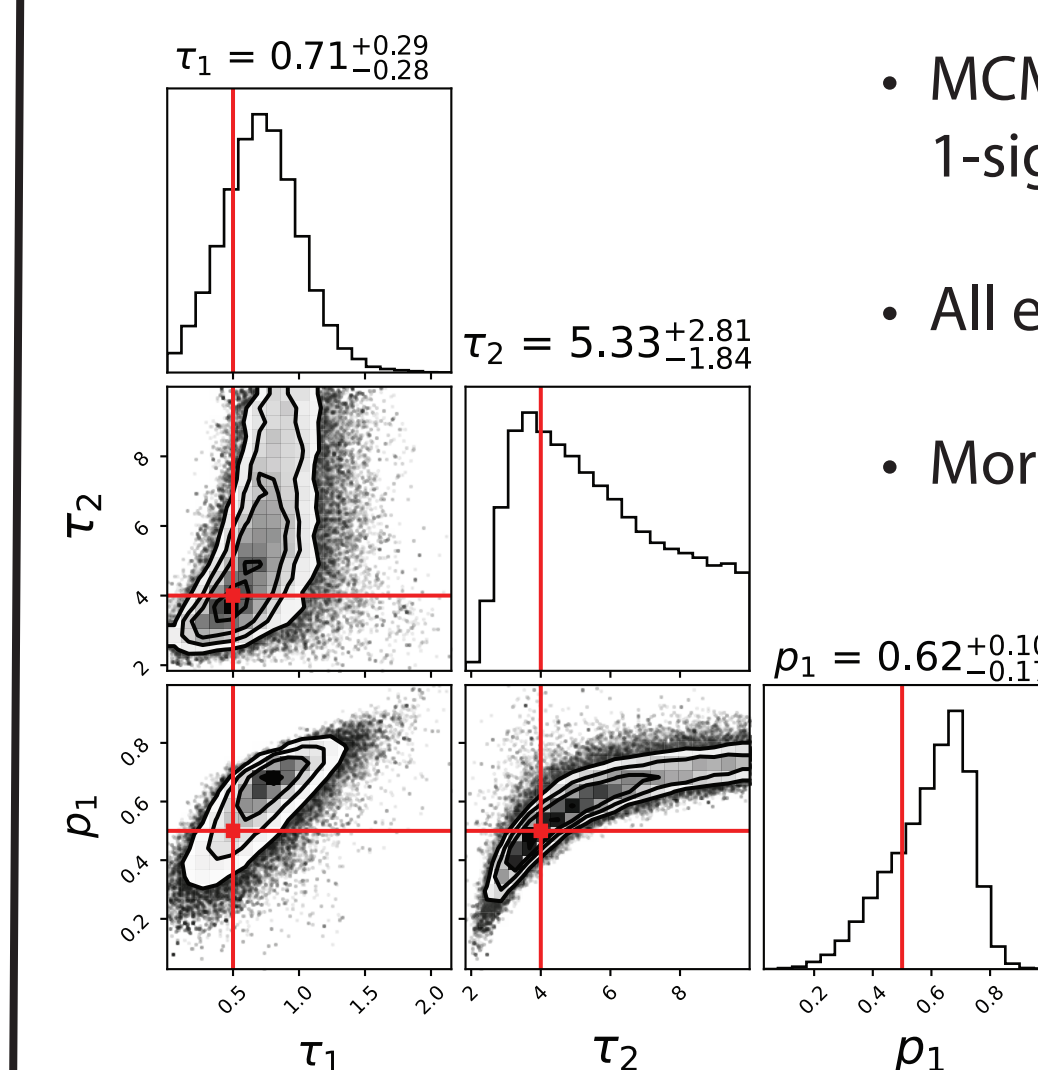
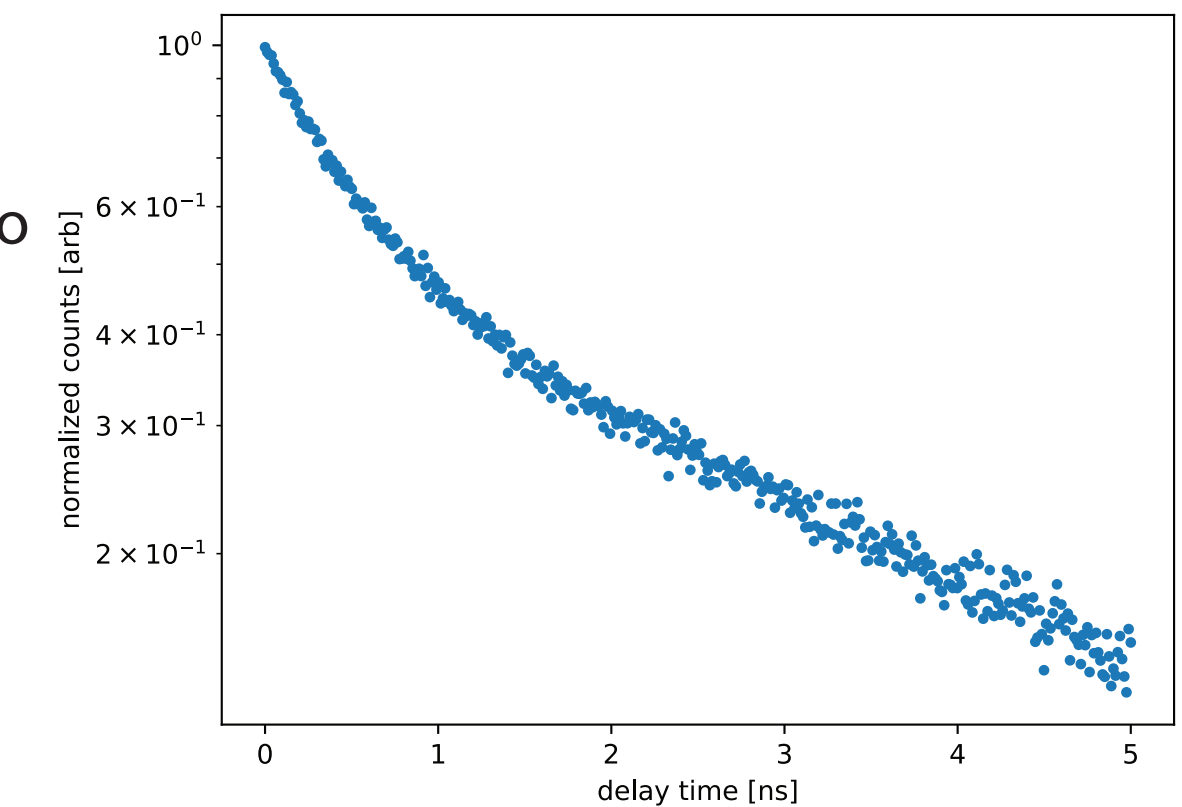
- Initial guess:
$$\tau_1 = 1\text{ns}, \tau_2 = 5\text{ns}, p_1 = 0.3$$
- 100,000 iterations
- Jump magnitudes 0.05, 0.1, 0.01
- 33% acceptance rate



- All parameters identified within 1-sigma
- Non-elliptical contours indicate nonlinear covariance
- MCMC converges very quickly (200 sample burn-in)

Control: No convolution

- Simulated same parameters as previous, but with no convolution and a smaller noise magnitude (0.01)
- MCMC has same iterations and starting point
- Jump magnitude calibrated for 20-50% acceptance rate



- MCMC failed to identify p_1 within 1-sigma
- All error bars are much wider
- More nonlinearity in contours

Conclusions

- Gaussian convolution seems to make fitting easier - blurring effect?
 - MCMC can fit convolved data
 - Non-elliptical contours challenge conventional uncertainty estimation and require confidence regions
- Future work:
- Real experimental data
 - More realistic noise model- allow for experimental background
 - Comparison with existing likelihood minimization